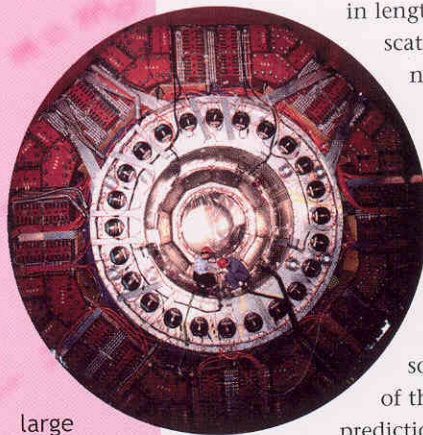


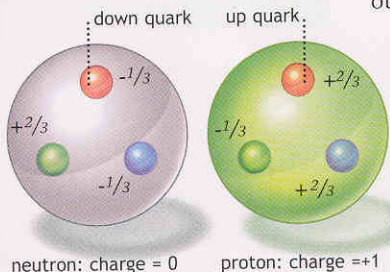


large detector

A giant solenoid magnet lies at the heart of the SLAC Large Detector; this device monitors the particles produced in the Linear Accelerator.

ups and downs

A neutron has two "down" quarks with a charge of $-1/3$, and an "up" quark, with a charge of $+2/3$, giving it no charge. A proton has two "up" quarks and one "down" quark, giving an overall charge of $+1$. "Up" and "down" are whimsical names given by physicists.



in length, and observing how they scattered from nucleons—protons or neutrons. The data showed that the electrons were bouncing off little hard particles within the nucleus. Richard Feynman called these particles "partons," a name chosen to avoid favoring any of the speculative models, and simply meaning that the particles are "part" of the nucleon. But it soon became clear that the behavior of these partons matched the predictions of quark theory exactly.

colors and glue

In its modern form, the quark model says that each nucleon is composed of three quarks (taken from "Three quarks for Muster Mark" in James Joyce's *Finnegans Wake*). The quarks have a property analogous to electric charge, which occurs in three varieties, not two (positive and negative). These varieties are labeled using the names of colors—purely for convenience. Quarks are not really colored, and the labels could have been Tom, Dick, and Harry. The names of the color charges are actually red, blue, and green. There is always one quark of each color in a nucleon, and the three colors in effect cancel each

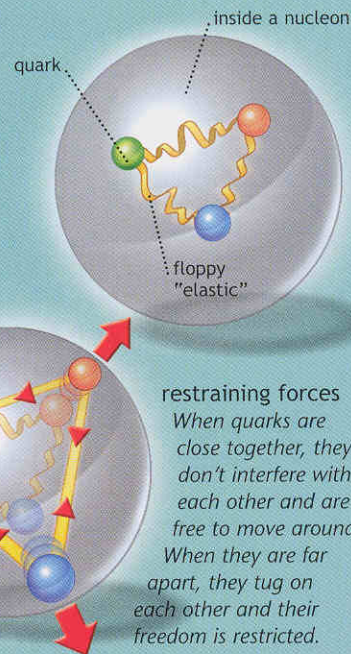
other out so that they have only a slight influence on other particles. However, the quarks can have electric charge, too. Just as the electromagnetic forces between

asymptotic freedom

Quarks inside a nucleon (neutron or proton) are confined as if they were held together by lengths of immensely strong elastic. When the quarks are close together, the elastic is floppy and they move freely. This is called asymptotic freedom. If you try to pull them apart, the elastic stretches, and the harder you pull, the more strongly they are pulled back, and so a single quark cannot exist on its own.

force (such as the impact of a particle) pulls quarks outward

"elastic" tightens and pulls quarks back toward each other



charged particles are carried by photons, so the color forces between quarks are carried by particles analogous to photons, whimsically called "gluons" because they hold quarks together, and this glue is very strong. Whereas the electric force gets weaker when charged particles are further apart, the glue force gets stronger as the two quarks are pulled further apart in the same way that it becomes harder to stretch a piece of elastic the more it's stretched. The result is that no quark can ever escape from the nucleon to have an independent existence of its own.